
Introductory Statistics for Econometrics

Unit 4. Expectation, Variance and Higher Moments

Mean and median, linearity, Jensen's inequality,
variance, standard deviation, skewness and kurtosis

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Learning outcomes. After this unit, the student should be able to (i) compute and interpret the expected value of a random variable; (ii) use the linearity of expectation as a calculation tool; (iii) apply LOTUS for $\mathbb{E}[g(X)]$; (iv) state and use Jensen's inequality, including its economic interpretation as risk aversion; (v) compute variance and standard deviation, and use the formulas for linear combinations under independence; (vi) define higher moments (skewness, kurtosis).

Source material: condensed from MIT OCW 14.30 (Spring 2009), K. Menzel, Lecture Notes 11–12.

1 Expected value

A full distribution may have far more information than we need. The *expected value* captures the centre of a distribution in a single number.

Definition 1.1 (Expected value). For a discrete random variable X with p.m.f. f_X ,

$$\mathbb{E}[X] = \sum_x x f_X(x),$$

provided the sum converges absolutely. For a continuous X with p.d.f. f_X ,

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx,$$

provided the integral converges absolutely. If the sum/integral fails to converge, $\mathbb{E}[X]$ does not exist.

The expected value is generally not a “typical” value. Rolling a fair die gives $\mathbb{E}[X] = 3.5$, an outcome that never occurs. The expectation is a centre of mass, not a representative value.

Example 1.2 (Bernoulli and binomial via indicators). If $X \sim \text{Bernoulli}(p)$ then $\mathbb{E}[X] = p$. If $Y \sim B(n, p)$ we can write $Y = \sum_{i=1}^n X_i$ with X_i i.i.d. Bernoulli(p). By the linearity of expectation (Property 3 below),

$$\mathbb{E}[Y] = \sum_{i=1}^n \mathbb{E}[X_i] = np.$$

This is much faster than evaluating $\sum_{x=0}^n x \binom{n}{x} p^x (1-p)^{n-x}$ directly.

2 Properties of expectation

Proposition 2.1 (Algebra of expectation). *Let X, Y be random variables and $a, b \in \mathbb{R}$.*

- (1) Constants: $\mathbb{E}[c] = c$ for any constant c .
- (2) Affine: $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$.
- (3) Linearity: $\mathbb{E}[a_1X_1 + \dots + a_nX_n + b] = \sum_i a_i \mathbb{E}[X_i] + b$, with no requirement of independence.
- (4) Multiplication under independence: if $X \perp Y$ then $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$. Without independence, this is generally false.

Theorem 2.2 (LOTUS — law of the unconscious statistician). *For a function $g : \mathbb{R} \rightarrow \mathbb{R}$,*

$$\mathbb{E}[g(X)] = \begin{cases} \sum_x g(x) f_X(x), & X \text{ discrete,} \\ \int_{-\infty}^{\infty} g(x) f_X(x) dx, & X \text{ continuous.} \end{cases}$$

The bivariate analogue is $\mathbb{E}[g(X, Y)] = \iint g(x, y) f_{XY}(x, y) dx dy$.

Remark 2.3 (Why LOTUS matters). LOTUS lets us compute $\mathbb{E}[g(X)]$ without first deriving the distribution of $g(X)$. The variance below, the moment-generating function, and most expected utilities in economics are computed this way.

Example 2.4 (Two LOTUS calculations). (a) If $f_X(x) = 2x$ on $(0, 1)$ and $Y = \sqrt{X}$, then $\mathbb{E}[Y] = \int_0^1 \sqrt{x} \cdot 2x dx = \frac{4}{5}$. (b) If $f_{XY} \equiv 1$ on $(0, 1)^2$ and $Z = X^2 + Y^2$, then $\mathbb{E}[Z] = \int_0^1 \int_0^1 (x^2 + y^2) dx dy = \frac{2}{3}$.

3 Median and quantiles

The expected value is one of several reasonable centres of a distribution; the *median* is another.

Definition 3.1 (Median, quantile). The *median* of X is $\text{med}(X) := \min\{m \in \mathbb{R} : \mathbb{P}(X \leq m) \geq \frac{1}{2}\}$. More generally, the α -*quantile* is $q(X, \alpha) := \min\{q \in \mathbb{R} : \mathbb{P}(X \leq q) \geq \alpha\}$, $\alpha \in (0, 1)$. The 50th percentile is the median; the 25th, 50th and 75th are the *quantiles*.

Example 3.2 (Mean $\neq$ median). For $f_X(x) = x^2/9$ on $(0, 3)$, $\mathbb{E}[X] = \int_0^3 x \cdot x^2/9 dx = 9/4 = 2.25$, while $F_X(x) = x^3/27$ gives $\text{med}(X) = \sqrt[3]{27/2} = \sqrt[3]{13.5} \approx 2.38$. The right-skew shifts the mean below the median; for left-skewed distributions the inequality reverses.

Proposition 3.3 (Median and monotone transforms). *If u is strictly increasing on the support of X , then $\text{med}(u(X)) = u(\text{med}(X))$.*

Two cautions.

- *The median is not linear:* in general $\text{med}(X + Y) \neq \text{med}(X) + \text{med}(Y)$.
- *Quantiles of averages $\neq$ averages of quantiles:* being above the 90th percentile in each component of a vector test does not put you above the 90th percentile in the total.

4 Jensen's inequality

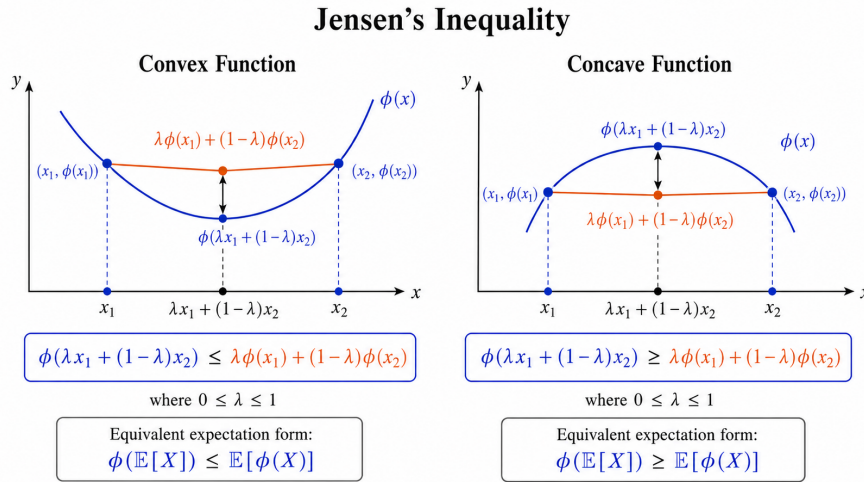
For affine u , $\mathbb{E}[u(X)] = u(\mathbb{E}[X])$. For non-linear u , the two are no longer equal — and they are systematically ordered when u is convex (or concave).

Theorem 4.1 (Jensen's inequality). Let $u : \mathbb{R} \rightarrow \mathbb{R}$ be a convex function and X a random variable with $\mathbb{E}[X] < \infty$. Then

$$\mathbb{E}[u(X)] \geq u(\mathbb{E}[X]). \quad (1)$$

The inequality is strict if u is strictly convex and X is non-degenerate. For concave u , the inequality reverses.

Sketch. At the point $\mathbb{E}[X]$ the convex function u lies above its tangent line $r(x) = u(\mathbb{E}[X]) + u'(\mathbb{E}[X])(x - \mathbb{E}[X])$. Taking expectations of $u(X) \geq r(X)$ and using linearity gives (1). $\square$



Economic content: risk aversion. A consumer with concave utility u (the standard assumption) prefers a sure outcome of $\mathbb{E}[W]$ to the lottery W itself: by Jensen, $u(\mathbb{E}[W]) \geq \mathbb{E}[u(W)]$. The amount one would pay to avoid the lottery is the *risk premium*, and is positive whenever u is strictly concave.

Example 4.2 (Should you buy the extended warranty?). A laptop costs \$1,200; with probability $p = 0.10$ a repair costs an extra \$1,100, otherwise nothing. An extended warranty covers all repairs for a flat \$115. With initial wealth \$4,800 and utility $u(c) = \sqrt{4,800 - c}$ (mildly concave),

$$\mathbb{E}[u(C_{\text{no plan}})] = 0.9\sqrt{3,600} + 0.1\sqrt{2,500} = 59, \quad u(C_{\text{plan}}) = \sqrt{3,485} \approx 59.04 > 59.$$

The risk-averse consumer buys the plan even though its expected cost (\$115) exceeds the expected loss without it (\$110). The \$5 gap is the *risk premium*.

Example 4.3 (The St. Petersburg paradox). A fair coin is tossed until the first head; you win $\$2^x$ if the first head occurs on toss x . Expected winnings: $\mathbb{E}[Y] = \sum_{x=1}^{\infty} 2^x \cdot 2^{-x} = \infty$. Yet most people will pay only a few dollars to play. Explanation: utility is concave (diminishing marginal value of money), so $\mathbb{E}[u(Y)] < \infty$ even when $\mathbb{E}[Y] = \infty$. Jensen's inequality reconciles theory and behaviour.

5 Variance and standard deviation

The expected value tells us where the distribution is centred; the variance tells us how spread out it is.

Definition 5.1 (Variance, standard deviation).

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2], \quad \sigma(X) = \sqrt{\text{Var}(X)}.$$

Proposition 5.2 (Properties of variance). (1) Computational formula: $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.

(2) Affine: $\text{Var}(aX + b) = a^2 \text{Var}(X)$.

(3) Independence: if $X_1, \dots, X_n$ are independent, $\text{Var}(\sum_i a_i X_i) = \sum_i a_i^2 \text{Var}(X_i)$.

(4) Degeneracy: $\text{Var}(X) = 0$ iff $\mathbb{P}(X = c) = 1$ for some constant c .

Example 5.3 (Variance of a binomial). For $Y \sim B(n, p)$ written as $Y = \sum_i X_i$ with X_i i.i.d. Bernoulli(p), $\mathbb{E}[X_i] = p$ and $\mathbb{E}[X_i^2] = p$, so $\text{Var}(X_i) = p - p^2 = p(1 - p)$. By independence $\text{Var}(Y) = np(1 - p)$.

Example 5.4 (Useful drill). Let X be uniform on $\{-2, 0, 1, 3, 4\}$. Compute $\mathbb{E}[X] = 6/5$, $\mathbb{E}[X^2] = 6$, $\text{Var}(X) = 6 - (6/5)^2 = 114/25$. For $Y = 4X - 7$, $\text{Var}(Y) = 16\text{Var}(X) = 1824/25 \approx 72.96$.

Remark 5.5 (Why we centre by the mean). Among all constants c , the one minimising $\mathbb{E}[(X - c)^2]$ is $c = \mathbb{E}[X]$. Variance is the smallest mean-squared deviation around any constant; this is the optimisation reason why “expected value” is the right notion of centre for L^2 analysis. We will reuse this fact when deriving the OLS estimator in Unit 9.

6 Higher moments: skewness and kurtosis

Definition 6.1 (Moments). The r th raw moment is $\mu'_r = \mathbb{E}[X^r]$. The r th central moment is $\mu_r = \mathbb{E}[(X - \mathbb{E}[X])^r]$.

So the mean is the first raw moment, the variance is the second central moment. Two further central moments are commonly used.

Skewness. The standardised third central moment, $\gamma_1 = \mu_3/\sigma^3$, measures asymmetry. $\gamma_1 > 0$ indicates a long right tail (right skew, common for incomes); $\gamma_1 < 0$ a long left tail; symmetric distributions have $\gamma_1 = 0$.

Kurtosis. The standardised fourth central moment, $\gamma_2 = \mu_4/\sigma^4$, measures tail weight. The normal distribution has $\gamma_2 = 3$; *excess kurtosis* is $\gamma_2 - 3$. Positive excess kurtosis (“fat tails”) is the empirical norm for asset returns and is the basis for the criticism of Gaussian risk models that became famous in the 2008 financial crisis.

7 Order statistics (a brief look)

Definition 7.1 (Order statistic). For an i.i.d. sample $X_1, \dots, X_n$ with c.d.f. F_X and p.d.f. f_X , the k th order statistic Y_k is the k th smallest value. In particular $Y_n = \max_i X_i$ and $Y_1 = \min_i X_i$.

Proposition 7.2 (Density of an order statistic).

$$f_{Y_k}(y) = k \binom{n}{k} F_X(y)^{k-1} [1 - F_X(y)]^{n-k} f_X(y).$$

In particular, $f_{Y_n}(y) = n F_X(y)^{n-1} f_X(y)$ and $f_{Y_1}(y) = n [1 - F_X(y)]^{n-1} f_X(y)$.

Example 7.3 (Auctions). n bidders submit i.i.d. bids B_i with c.d.f. F_B . In a *first-price* (Dutch) auction the seller’s revenue is the maximum bid, with density $n F_B(y)^{n-1} f_B(y)$. In a *second-price* (English) auction the seller’s revenue is the second-highest bid, with density $n(n-1) F_B(y)^{n-2} [1 - F_B(y)] f_B(y)$. These distributions are the building blocks of empirical auction theory in industrial economics.

Summary

- Expected value $\mathbb{E}[X] = \sum_x x f_X(x)$ or $\int x f_X(x) dx$.
- Linearity: $\mathbb{E}[\sum_i a_i X_i + b] = \sum_i a_i \mathbb{E}[X_i] + b$, no independence needed; multiplicativity $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ requires $X \perp Y$.
- LOTUS: $\mathbb{E}[g(X)] = \int g f_X$.
- Median and quantiles capture order; medians commute with strictly increasing transformations but not with sums.
- Jensen's inequality: $\mathbb{E}[u(X)] \geq u(\mathbb{E}[X])$ for convex u ; concave u reverses. Risk aversion = concave utility.
- Variance: $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$, $\text{Var}(aX + b) = a^2 \text{Var}(X)$, and additive over independent summands.
- Higher central moments: skewness measures asymmetry, kurtosis tail weight.

Suggested exercises

1. Let X have p.d.f. $f_X(x) = 2x$ on $(0, 1)$. Compute $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, $\text{Var}(X)$, and $\text{med}(X)$.
2. For $X \sim B(n, p)$, derive $\mathbb{E}[X(X-1)] = n(n-1)p^2$ and use it to give an alternative proof that $\text{Var}(X) = np(1-p)$.
3. Show that among all constants c , $\mathbb{E}[(X-c)^2]$ is minimised at $c = \mathbb{E}[X]$. (Differentiate or complete the square.)
4. (Jensen.) Let X be positive with $\mathbb{E}[X] = \mu$. Use Jensen to show $\mathbb{E}[\log X] \leq \log \mu$ and $\mathbb{E}[1/X] \geq 1/\mu$. Interpret.
5. (Risk aversion.) Repeat Example 4.2 for general $u(c) = \sqrt{W_0 - c}$ and find the maximum K such that the consumer buys the plan for any premium less than K .
6. (Order statistics.) Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} U[0, 1]$. Show that $\mathbb{E}[Y_k] = k/(n+1)$ and that the density of Y_k is the $\text{Beta}(k, n-k+1)$ density.